

```

*** - EVAL: undefined function F00
The following restarts are available:
USE-VALUE      :R1   Input a value to be used instead of
(FDEFINITION 'F00).
RETRY          :R2   Retry
STORE-VALUE    :R3   Input a new value for (FDEFINITION
'F00).
ABORT          :R4   Abort main loop
>

```

That's right! You get slapped in the face with an error (Figure 3). The *flet* command creates local functions, which are invisible inside another function body.

To define recursive local functions, use *labels*. Local functions defined by a labels expression can refer to any other functions defined there, including themselves. This is identical in its basic structure to the *flet* command. Here's the preceding code, written with *labels*:

```

> (labels ( (foo(n)
              (+ n 5))
            (bar(n)
              (+ (foo 2) n)))
  (bar 10))

```

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In this code example, the body of the *labels* expression calls the local function *bar*. Since *labels* allows local functions to 'see' each other, the *bar* function calls another local function (from inside its body). That is *foo*, which adds 2 to 5, resulting in a value of 7, which in turn is added to 10—the argument of the *bar* function, resulting in a final value of 17.

This is where I have to leave you this week (as I have run out of space!) but before I go, I would like to thank everybody who wrote in to provide feedback about the article last month. **END** 

By: Vivek Shangari

The author is a hard-core hacker, if there ever was one. He always thinks programs, and holds a piece of code in his head all the time. His favourite past time is to stop random strangers on the street and start talking about the benefits of open source over proprietary software, till they agree to switch sides, or threaten to jump off the nearest building. Every now and then, he manages to talk a bunch of unsuspecting potential programmers into letting him train them in Lisp. It's not rare to find people stumble out after the training session 'zombified', or with tears running down their faces. "Those are tears of joy!" he will patiently explain. For those of you out there who cannot be deterred, you can reach him at vivek@codeCraft.in.

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suggestion or a complaint?

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